MEASUREMENT OF SEGMENTS

5–1 INTRODUCTION

In this chapter we are going to get a number for the length of a segment, that is for the distance between two points. The way we get the length will be exactly the same as the method we use in practical work. We will see that finding a length is based upon simple *counting*. For example, when you pace off the length of a building (Fig. 5–1), you count how many steps you take. Or when you say that the room is 15 feet wide, you mean that a foot ruler can be laid down 15 times, one after the other, on a line. If the width of the room is between 14 and 15 feet, the ruler can be laid down 14 times, but 15 times would put the end of the ruler past the wall. To measure the room more accurately, you must choose a smaller unit of measure, say an inch, so that if the width is 14 feet and three inches, the foot ruler can be laid down 14 times and beyond that the inch ruler can be laid down three times. If even then you have not come exactly to the wall of the room, you must choose an even smaller unit of measure, some fraction of an inch, and so on.

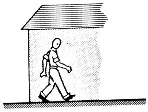

FIGURE 5–1

Exercise Group 5–1

1. Measure the length of your desk. Describe the process of measurement in terms of counting.
2. Measure the length of a page. Describe the counting you do to get the length.
3. Think of other examples of measurement and describe them in terms of counting.

5–2 NUMBERS

Because we express length in terms of numbers, we need certain facts about numbers that you learned in algebra and arithmetic. You have studied different kinds of numbers. In the beginning grades you first used whole numbers 1, 2, . . . and learned how to add, subtract, and multiply them. These numbers we now call the *positive integers* instead of "whole numbers." When you tried to divide integers you found that it was necessary to have other numbers, which you called *fractions*, for example, $\frac{1}{2}$, $\frac{2}{3}$, $\frac{5}{8}$, $\frac{7}{37}$. These are numbers obtained by dividing one integer by another integer. *The numbers which are either positive integers or positive fractions are called positive rational numbers.*

You also wrote numbers as *decimals*, for example, 1.25, 2.537, 173.612. Sometimes a rational number when written as a decimal "comes out even," as for example, $\frac{1}{2} = 0.5$, $\frac{3}{8} = 0.375$. Sometimes it does not come out even, so that the decimal goes on forever, for example, $\frac{1}{3} = 0.333\ldots$, $\frac{2}{3} = 0.6666\ldots$, $\frac{1}{11} = 0.090909\ldots$. You also met with other numbers which did not come out even, for example, $\sqrt{2} = 1.4142\ldots$, $\sqrt{3} = 1.7321\ldots$. These numbers are different from $\frac{1}{3}$, $\frac{2}{3}$, and $\frac{1}{11}$ because they not only do not come out even, but *the decimal never repeats*. In other words, no matter to how many decimal places you were to calculate $\sqrt{2}$, you would never reach a place where, from then on, one group of digits would occur over and over again in the same order.

Decimals that, after some place, repeat one group of digits over and over are called *repeating decimals*. Decimals which come out even are called *terminating decimals*, and may be considered as repeating with zeros.

Exercise 5–2

Write the following rational numbers as decimals to see that they are repeating decimals. If they come out even (terminate), the repeating part is all zeros. 1/6, 5/6, 122/99, 1/13, 1/7, 6/5, 3/2.

It is proved in advanced algebra that rational numbers are always repeating decimals, and that, conversely, repeating decimals are rational numbers. (These proofs are hard, so don't be discouraged if you cannot construct them; your teacher may show you.) Hence, rational numbers and repeating decimals are the same. What, then, about the decimals which never repeat? These numbers are not rational numbers, and so they have been called *irrational numbers*.* Irrational numbers are defined to be

* The word "irrational" is not a very pleasant choice because it sounds like "not sensible." What it really means is this. Rational numbers are obtained by dividing one integer by another; that is, a rational number is the ratio of two integers. Hence the word "rational" in "rational number" means that the number is a *ratio* of integers. Therefore, an *irrational* number is one which is *not* the ratio of two integers.

those numbers which are not rational. Examples of irrational numbers are $\sqrt{2}$, $\sqrt{3}$, $\sqrt{5}$, $\sqrt[3]{2}$, $\sqrt[3]{17}$, and, in fact, any root of an integer which is not itself an integer.

All the decimals together are defined to be the *real numbers*. Thus a real number is a decimal. If the decimal repeats, the number is a rational number. If the decimal does not repeat, it is an irrational number.

Exercise Group 5–3

1. Write the symbols for several rational numbers and for several irrational numbers.
2. Is 3.131313..., where the 1's and 3's keep alternating, a rational number?
3. Is 0.234234234..., where 2, 3, and 4 keep following each other, a rational number?

*4. The number 1.010010001... is formed from just 0's and 1's. After each 1 there is one more 0 than the previous time. Is this number rational?

5. Give the definitions of rational number, irrational number, real number.
6. A real number is either rational or irrational. Why?

*7. There is a rule for deciding when a fraction written as a decimal will have its repeating part all zeros and when it will not. For example, each of the fractions $3/20$, $17/250$, $5/16$, and $4/25$ has its repeating decimal part all zeros. The fractions $7/15$, $5/12$, and $11/23$ do not. Can you discover the rule?

When we write a decimal, for example 1.23, we mean $1 + 0.2 + 0.03$ or $1 + 2/10 + 3/100$. Similarly, $4.383 = 4 + 3/10 + 8/100 + 3/1000$. Thus, any terminating decimal (one which comes out even) may be written as a sum of ordinary fractions with denominators 10, 100, ... If the decimal is repeating, this sum is an *infinite* one, for example, $1/3 = 0.333\ldots = 3/10 + 3/100 + 3/1000 + 3/10000 + \cdots$.

Exercise Group 5–4

1. Write the following decimals as sums of rational numbers: 4.82, 1.29, 2.75, 7.122, 0.8875, 2.40. In order to express each number as a single fraction, add the rational numbers.
2. Write the following numbers as decimals: $3 + 2/10 + 4/100$, $2 + 3/10 + 5/100 + 6/1000$, $4 + 3/100$, $1 + 1/10 + 1/10000$.
3. Write the following repeating decimals as sums of rational numbers: 0.6666..., 0.7777..., 0.121212....
4. Write the following numbers as decimals: $1 + 2/10 + 2/100 + 2/1000 + \cdots$, $3 + 1/10 + 4/100 + 1/1000 + 4/10000 + \cdots$, $8/10 + 8/100 + 8/1000 + 8/10000 + \cdots$.

5–3 INEQUALITIES

In this book you are expected to know how to do arithmetic, that is, how to add, subtract, multiply, and divide real numbers. But there is one part of arithmetic you have not yet studied systematically. That part is concerned with comparing the sizes of real numbers. Since you already know what it means to say that the number x is larger than the number y, all we need is a symbol to indicate this. We therefore make the definition: $x > y$ means "x is greater than y." Another way to say this is "$x - y$ is a positive number." We also define $x < y$ to mean "x is less than y," or, in other words "y is greater than x." Listed below are several properties of real numbers. These properties can be established as indicated in the review of Chapter 4.

(a) If $x > y$ and $y = z$, then $x > z$.

(b) If $x > y$ and $y > z$, then $x > z$.

(c) x is a positive number if and only if $x > 0$.

(d) x is a negative number if and only if $x < 0$.

(e) If $x > y$, then $x - y > 0$, and conversely.

We define $x \geqq y$ to mean "either $x > y$ or $x = y$," and we read "$\geqq$" as "is greater than or equal to." We make a similar definition for $x \leqq y$ ("x is less than or equal to y").

(f) If $x > y$ and $u \geqq v$, then $x + u > y + v$.

Exercise Group 5–5

1. Substitute numbers for the variables in properties (a), (b), (e), and (f) above, and thus verify their truth for certain cases.
2. Write in symbols: a is greater than b; a is greater than or equal to b; u is less than w; $a + c$ is less than or equal to $r + s$; $(a + b)^2$ is positive; z is a negative number.
3. If x is any real number, $x^2 \geqq 0$. Why?
4. Why is it true that if $x > 0$ and $y > 0$, then $xy > 0$?
5. Tell why $xy < 0$ if $x > 0$ and $y < 0$.

5–4 MEASUREMENT

To measure a line segment we must first tell what our unit of length is. Some units of length often used in practical work are inch, foot, yard, mile, meter, centimeter, etc. Having picked a unit of length, we will describe the geometric process whereby we get a real number, that is, a decimal for the length of a line segment.

Choose any segment CD as a unit (Fig. 5–2). This segment and all segments congruent to it are defined to have length 1. Suppose AB is any segment on a line l. On the same side of A as B there is a point A_1 such that $AA_1 \cong CD$. If $A_1 = B$, then the length of AB is 1.

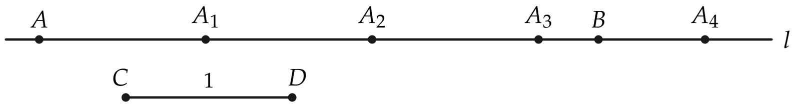

FIGURE 5–2

FIGURE 5–3

If B lies between A and A_1 (Fig. 5–3), then our unit was too large to measure the length of AB in integers (whole numbers). We will return to this possibility later. If A_1 lies between A and B, then there is a point A_2, on the same side of A_1 as B, such that either $A_2 = B$, or B is between A_1 and A_2, or A_2 is between A_1 and B. If $A_2 = B$, the length of AB is 2. If B is between A_1 and A_2, then our unit will not measure the length in integers. If A_2 is between A_1 and B, there is a point A_3, on the same side of A_2 as B, such that $A_2A_3 \cong CD$.

The process described above is what we actually follow when we measure. If we lay out segments congruent to CD from A toward B, we know from experience that we eventually either reach B exactly, so that the length of AB is an integer, or reach a point *beyond* B. Thus in Fig. 5–2, A_4 is beyond B but A_3 is still between A and B. The length in this case will be a number between 3 and 4. If the above process can always be carried out in our geometry, then we have described the beginning of the process of measurement.

The only part of the above process that does not follow from our postulates is the proof that we can always get to or beyond B. Because we need to do this, we now add a new postulate to our list. It was the Greek mathematician Archimedes (287–212 B.C.) who first stated this postulate. He apparently did not see how to get it from Euclid's postulates and, needing it, assumed it was true. It is one of the things that Euclid forgot to state as a postulate. It is called *Archimedes' axiom* or *Archimedes' postulate* in spite of its use by Eudoxus much earlier. This is the first of the two new postulates in Postulate Group IV which we need for measurement. The other one will be given later.

IV–1 *Archimedes' postulate.* If AB is any segment on a line l and CD any other segment, then there is a definite number n of points $A_1, A_2, \ldots, A_n$ on l such that the points $A = A_0, A_1, A_2, \ldots, A_n$ are arranged in this order, and all the segments $AA_1, A_1A_2, \ldots, A_{n-1}A_n$ are congruent to CD, and either $B = A_n$ or B lies between A_{n-1} and A_n.

The integer n is called *Archimedes' integer*. When n is 4, an illustration would look like Fig. 5–4 or Fig. 5–5.

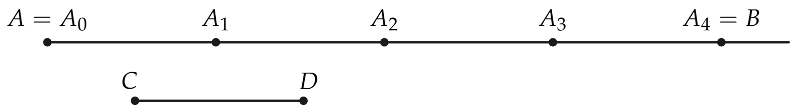

FIGURE 5–4

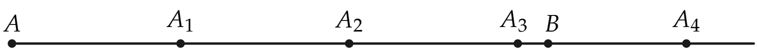

FIGURE 5–5

Exercise Group 5–6

1. If AB is this segment A——B and CD is this segment C—D, find how many points are needed to get beyond B.
2. Draw a segment AB and a segment CD. Find the integer n of Archimedes' axiom for your segments.

Using Archimedes' postulate, we find that the length of AB is either an integer or is between two integers. If the length lies between two integers, the next step in the process is to divide the unit interval into tenths (Fig. 5–6). (We would not have to keep to tenths if we were not trying to come out with a decimal.) Then we count "tenths" until either we reach B exactly or B lies between two of the tenth marks. If we reach B exactly, then the length of AB comes out exactly in tenths. If B is not reached exactly, then we must divide the unit interval into one-hundredths and repeat the process. In this way we obtain a real number or decimal, for the length of AB. In Fig. 5–7 the length of AB is about 2.3.

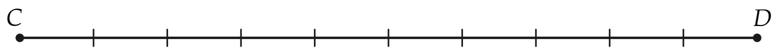

FIGURE 5–6

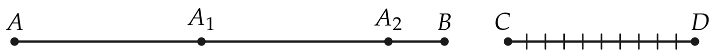

FIGURE 5–7

We have indicated a segment by writing the two endpoints of the segment as AB. At this point we also need a notation for the *length* of the segment. For this we will use $\overline{AB}$, that is $\overline{AB}$ = *the length of the segment AB*. Whenever you see the symbol $\overline{AB}$ you must remember that it represents a *number*, while the symbol AB stands for a *set of points*.

Except for dividing a segment into tenths, the entire procedure of this section can be carried out (that is, described) in our geometry. At the moment we do not have the means for dividing a segment into tenths. As soon as we have studied parallel lines (Chapter 8), we will be able to make this construction. Since we do not need the results of this chapter until after we have studied parallel lines, our procedure is logically correct.

Exercise Group 5–7

1. Draw a line segment. Then draw a segment that will be your unit. Now measure the first segment, describing how you use Archimedes' postulate and relations of betweenness in the process.
2. Draw a line segment. Measure it with your ruler to the nearest 1/16 of an inch. Describe your measurement process, using Archimedes' postulate and betweenness.

5–5 POINTS ON A LINE

We have just described the measurement process for getting a number for the length of a segment. We now turn the question around and ask whether, given a number, there is a segment with that number for its length. Of course, we would like this to be true, but our postulates so far do not tell us that it is true. The process of measurement describes a method for "sneaking up" on the point B, but if we start out with a number we are not assured that there is a point to sneak up on! To have this property, we must postulate it. This postulate tells us that there are enough points on the line.

IV–2. *The completeness postulate.* For every line l and any point A on l, and for any positive real number x, there is, on a given side of A, a point B such that $\overline{AB} = x$.

The name of the last postulate is meant to signify that our lines are complete, that is, that they have all the points on them we expect them to have.

Exercise Group 5–8

1. Draw a segment that is to be the unit segment. Draw another segment that is 1.3 units in length.
2. With the unit segment of Exercise 1, draw a segment of length $2 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8}$ units.

5–6 LENGTH AND CONGRUENCE

When we started talking about congruent segments we had in mind a notion of size. Now that we have lengths for segments it should turn out that congruent segments have the same length, and conversely that segments of the same length are congruent. We now prove these theorems. To do so we need the following theorem (4–3) that we proved in Chapter 4. *If $AB \cong A'B'$ and ACB and if C' is on the same side of A' as B' and $AC \cong A'C'$, then $A'C'B'$.*

***Theorem 5–1.** *If $AB \cong A'B'$, then $\overline{AB} = \overline{A'B'}$.*

Proof. To get the length of segment AB, we establish a sequence of points *between* A and B. Let us designate this sequence by $C_1, C_2, C_3, \ldots$. Now, consider the corresponding sequence, C_1', C_2', $C_3', \ldots$, where $AC_1 \cong A'C_1'$, $AC_2 \cong A'C_2'$, $AC_3 \cong A'C_3', \ldots$, and all the points $C_1', C_2', C_3', \ldots$ are on the same side of A' as B'. Using Theorem 4–3, we see that all the points $C_1', C_2', C_3', \ldots$, are between A' and B'; therefore, this sequence determines the same real number (decimal) as does the sequence $C_1, C_2, C_3, \ldots$.

***Theorem 5–2.** *If* $\overline{AB} = \overline{A'B'}$, *then* $AB \cong A'B'$.

Proof. This theorem is the converse of Theorem 5–1. We will prove it by choosing a point B'', on the same side of A as B, such that $AB'' \cong A'B'$, and then proving that $B'' = B$ (Fig. 5–8). This will show that $AB \cong A'B'$. Either $B'' = B$ or $B'' \neq B$. Since $AB'' \cong A'B'$, then from Theorem 5–1 $\overline{AB''} = \overline{A'B'}$. We are given that $\overline{AB} = \overline{A'B'}$. Hence $\overline{AB''} = \overline{AB}$. Thus we must show that if $B'' \neq B$, we cannot have $\overline{AB''} = \overline{A'B'}$. But if $B'' \neq B$, then, if the unit segment is smaller than BB'', the corresponding Archimedes integers for AB and AB'' would be different. This would lead to different lengths for AB and AB''. Hence B'' must be the same as B and $AB \cong A'B'$.

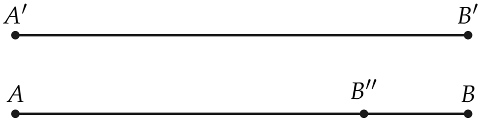

FIGURE 5–8

These two theorems give us the right to use the words "the same length" in place of "congruent." Therefore we can use everyday language in talking about congruence of segments.

Exercise Group 5–9

1. $AB \cong CD$ and $CD \cong EF$, and $\overline{AB} = 7.26$ inches. What is $\overline{EF}$?

2. If $\overline{PQ}$ is 1.23 ft, and $VU \cong RS$, and $\overline{RS} = 1.23$ ft, is $QP \cong VU$?

3. If $AB \cong CD$, $CD \cong EF$, $EF \cong GH$, is $\overline{AB} = \overline{GH}$?

4. An isosceles triangle can be defined now in two ways. State the two definitions.

5–7 MORE ABOUT INEQUALITIES

In Chapter 4 we defined $>$ and $<$ for segments. In this chapter (Section 5–3) we defined $>$ and $<$ for numbers. Since "same length" means "congruent," it should also be true that if $AB > CD$, then $\overline{AB} > \overline{CD}$, and conversely.

***Theorem 5–3.** *If $AB > CD$ then $\overline{AB} > \overline{CD}$, and conversely, if $\overline{AB} > \overline{CD}$ then $AB > CD$.*

Proof. The theorem could also be stated as "$AB > CD$ if and only if $\overline{AB} > \overline{CD}$." Or, again, as "If $AB > CD$ then $\overline{AB} > \overline{CD}$, and conversely." We first prove that $AB > CD$ implies $\overline{AB} > \overline{CD}$. The hypothesis, $AB > CD$, means that if E is on the same side of C as D, and $AB \cong CE$, then D is between C and E. (See Fig. 5–9.) By Theorem 5–1, $\overline{AB} = \overline{CE}$. In the measurement process, if the unit is chosen smaller than DE, the corresponding Archimedes' integers for CD and CE will be different, and in fact the integer for CD will be smaller than that for CE. Hence $\overline{CD} < \overline{CE}$ and, since $\overline{CE} = \overline{AB}$, $\overline{CD} < \overline{AB}$.

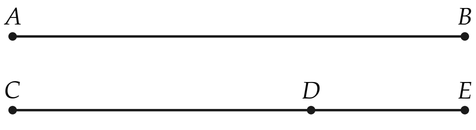

FIGURE 5–9

The converse is proved in a similar manner. We have given that $\overline{AB} > \overline{CD}$. Choose E, on the same side of C as D, such that $AB \cong CE$. By Theorem 5–1, $\overline{AB} = \overline{CE}$. Now either CDE, $D = E$, or CED. (Why?) We cannot have $D = E$. (Why?) An argument like the first part of the proof will show we cannot have CED.

Hence we must have CDE. It follows from the definition of "less than" for segments that $CD < CE \cong AB$.

Exercise Group 5–10

1. If ABC and $\overline{AB} = 5$, what can you say about $\overline{AC}$?
2. If $AB > EF$ and $\overline{EF} = 4$, what can you say about $\overline{AB}$?
3. If $AB > CD > EF$, what can you say about $\overline{AB}$ and $\overline{EF}$?
4. If $ABCD$, what can you say about $\overline{AD}$ and $\overline{BC}$?
5. If $\overline{AB} > \overline{AC}$, and C is a point on the line containing A and B, and C lies on the same side of A as B, where is the point C?

5–8 CHANGE OF SCALE

In measurement we choose a fixed segment for a unit of length. What happens if a different segment is chosen for the unit? This is a familiar situation because we are all used to dealing with inches, feet, yards, or miles to measure length. In Fig. 5–10, if CD is the unit, the length of AB is 3. If ED is the unit, the length of AB is 6. This illustrates the following theorem.

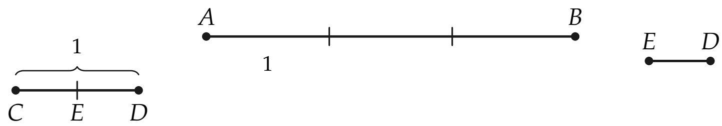

FIGURE 5–10

Theorem 5–4. *If $\overline{AB} = x$ when CD is the unit, and if $\overline{CD} = y$ when EF is the unit, then $AB = xy$ when EF is the unit.*

We will not give the proof of this theorem for arbitrary x and y because it is long to write out and quite hard if y is an irrational number. But if y is an integer, the proof is very easy and is really clear from Fig. 5–10, where $y = 2$.

Exercise Group 5–11

1. $\overline{AB} = 5$ when CD is the unit, and $\overline{CD} = 2.3$ when EF is the unit. What is $\overline{AB}$ when EF is the unit?
2. The width of the room is 14 ft. What is the width in inches? How does this follow from Theorem 5–4?
3. One kilometer is 1000 meters. One meter is 100 centimeters. How many centimeters in a kilometer? Show how this follows from Theorem 5–4.
4. $\overline{PQ} = 6.5$ when $\overline{RS} = 1$. $\overline{RS} = \frac{1}{2}$ when $\overline{UV} = 1$. What is $\overline{PQ}$ when $\overline{UV} = 1$?
5. If $CD < EF$, how does $\overline{AB}$ with unit CD compare with $\overline{AB}$ with unit EF? In other words, if we use two different units and compute two lengths for one segment, does the smaller unit give the smaller or the larger length?

REVIEW OF CHAPTER 5

1. In this chapter we have given definitions of the following: real number, rational number, irrational number, positive integer, greater than, less than, the process of measurement. Be sure you know these definitions.
2. This chapter has two new postulates, Archimedes' postulate and the completeness postulate. Go over them again and be sure you see what they mean.

3. In this chapter we have proved the following theorems. Study them again to be sure you know what they say. You should be able to state clearly the facts upon which each proof depends.

$$(AB \cong CD) \leftrightarrow \overline{AB} = \overline{CD},$$
$$(AB > CD) \leftrightarrow \overline{AB} > \overline{CD},$$
$$(AB < CD) \leftrightarrow \overline{AB} < \overline{CD}.$$

Note. In this chapter we have shown that congruence is equivalent to equality of length. From this point on in our development of geometry you may use the ordinary language of length in speaking of congruence.

ALGEBRA REVIEW

1. Segment XY has a length of 28 units. Point T is between X and Y and $\overline{XT} = \overline{YT} + 9$. Determine the lengths of XT and YT.

2. Points R, T, and S are in this order on a line. Segment RT is twice as long as segment TS. The difference in their lengths is 11. Find the length of RS.

3. Points A, B, and C are on one line. The length of AC is 12 more than that of AB. The length of BC is four times the length of AB. Find the lengths of AB and AC if (a) A is between B and C, and (b) B is between A and C.

4. Points A, B, C, and D are in this order on one line. $\overline{BC} = \overline{AB} + 4$; $\overline{CD} = \overline{AB} + \overline{BC}$; $\overline{AD} = 32$. Find $\overline{AB}$.

5. The length of XY is $\frac{2}{3}$ of the length of XZ. The length of XZ is $\frac{3}{5}$ of the length of YZ. If these three points are collinear: (a) What is the order of the points on l? (b) Express $\overline{XY}$ as a fraction of $\overline{YZ}$. (c) Determine $\overline{XY}$ if $\overline{XZ} - \overline{XY} = 6$.

6. The point M is the midpoint of segment AB, N is the midpoint of MB, and O is the midpoint of NB. If $\overline{AN} - \overline{NO} = 20$, find the length of AB.

7. $\overline{AC} = 2\overline{AB}$, $\overline{AB} + \overline{BC} = \overline{AC} + 2$. If $\overline{AB} > 1$, prove that the three points A, B, and C are not on one line.

8. A square and equilateral triangle have equal perimeters. The length of a side of the triangle exceeds the length of a side of the square by six inches. Find the perimeter of each.

9. The sum of the perimeters of a square and a rectangle is 36 ft. The rectangle is twice as long as it is wide, and the width of the rectangle is 1 ft more than the length of a side of the square. Find the dimensions of each.

10. The point B is the midpoint of segment AC. Point X is between B and C, and $\overline{BX} = 3$ inches. Show that a square having a side congruent to AB has an area of 9 in^2 more than a rectangle with sides congruent to AX and XC.